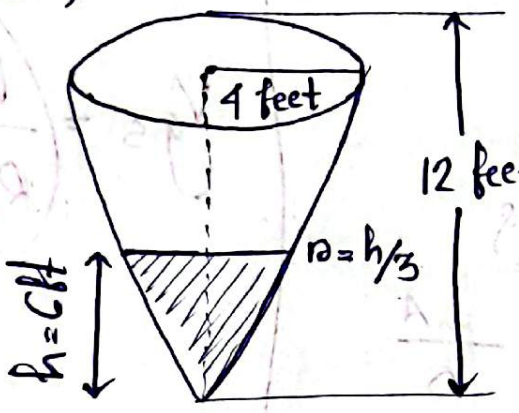
Assignment #3

Ans:no: 1

Given;



$\frac{dV}{dt} = 10\ ft^3/min$

Given; $h = 6\ ft$ $\qquad$ 1/(a)

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi\left(\frac{h}{3}\right)^2 h = \frac{1}{9}\pi h^3 \times \frac{1}{3} = \frac{1}{27}\pi h^3$$

$$\frac{dV}{dt} = \frac{\pi}{27} \times 3h^2 \times \frac{dh}{dt}$$

$$= \frac{\pi}{9} h^2 \frac{dh}{dt}$$

$$\Rightarrow \frac{dh}{dt} = \frac{\frac{dV}{dt}}{\frac{\pi}{9}h^2} = \frac{5}{\frac{\pi}{9} \times (6)^2}$$

$$= \frac{5}{2\pi}\ ft/min$$

$$= 0.795\ ft/min$$

∴ At $h = 6\ ft$;
the water level is rising at a speed of 0.795 feet per minute.

1/(b)

Lateral surface area, $A = \pi r \sqrt{r^2 + h^2}$; $[r = h/3]$

$$= \pi \left(\frac{h}{3}\right) \sqrt{\left(\frac{h}{3}\right)^2 + h^2}$$

$$= \pi \cdot \frac{h}{3} \sqrt{\frac{h^2}{9} + h^2} = \pi \times \frac{h}{3} \times \frac{h}{3} \times \sqrt{10}$$

$$= \frac{\sqrt{10}\,\pi h^2}{9}$$

$$\frac{dA}{dt} = \frac{\sqrt{10}}{9} \pi \times 2h \frac{dh}{dt}$$

From; (a)
$\frac{dh}{dt} = \frac{5}{2\pi}$ ft/min
$h = 6$ ft

$$= \frac{\sqrt{10}}{9} \times 2 \times 6 \times \frac{5}{2}$$

$$= \frac{10\sqrt{10}}{3} \text{ ft}^2/\text{min}$$

$$\approx 10.54 \text{ ft}^2/\text{min}$$

At 10.54 feet^2 per minute rate is the wetted surface area of the cone changing at the same instant.

Ans:

1/(c)

$\frac{dV}{dh} = ?$ | $h = 6$ feet, $r = \frac{h}{3}$

We know; $V = \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \pi \times \frac{h^2}{9} \times h = \frac{\pi}{27} h^3$

$\therefore \frac{dV}{dh} = 3 \times \frac{\pi}{27} h^2$

$= \frac{\pi}{9} h^2 = \frac{\pi}{9} \times 6^2 = 4\pi$ feet^3/feet

≈ 12.56 ft^3/ft

$\therefore \frac{dV}{dh} = 12.56$ ft^3/ft

Ans: no: 02

(a)

$$\lim_{x \to 0} \frac{e^{4x} - 1 - 4x}{x^2} \quad \left[\frac{0}{0}\right]$$

$$= \lim_{x \to 0} \frac{4e^{4x} - 4}{2x} \quad [\text{Using L' Hôpital}] \quad \left[\frac{0}{0}\right]$$

$$= \lim_{x \to 0} \frac{8}{\cancel{2}} \frac{16\, e^{4x}}{\cancel{x}} \quad [\text{Again using L' Hopital}]$$

$$= 8 \times e^0 = 8 \quad \text{Ans.}$$

(b)

$$\lim_{x \to 0} \frac{1 - \cos 3x}{x \cdot \tan x} \quad \left[\frac{0}{0}\right]$$

$$= \lim_{x \to 0} \frac{3 \sin 3x}{x \sec^2 x + \tan x} \quad \left[\frac{0}{0}\right]$$

$$= \lim_{x \to 0} \frac{9 \cos 3x}{x \cdot 2\sec x \cdot \sec x \tan x + 2\sec^2 x}$$

$$= \lim_{x \to 0} \frac{9 \cos 3x}{2x \sec^2 x \tan x + 2\sec^2 x}$$

$$= \frac{9 \cdot 1}{0 + 2} = \frac{9}{2}$$

2/(c)

$$\lim_{x \to \infty} \frac{\ln(x^3+1)}{x} \qquad \left[\frac{\infty}{\infty}\right]$$

$$= \lim_{x \to \infty} \frac{1}{x^3+1} \times 3x^2$$

$$= \lim_{x \to \infty} \frac{x^2(3)}{x^3\left(1+\frac{1}{x}\right)}$$

$$= \frac{3}{\infty} = 0$$

2/(d)

$$\lim_{x \to 0^+} x^2 \ln x \qquad [0 \cdot \infty]$$

$$= \lim_{x \to 0^+} \frac{\ln x}{1/x^2} \qquad \left[\frac{\infty}{\infty}\right]$$

$$= \lim_{x \to 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^4} \times 2x} = \lim_{x \to 0^+} \frac{\frac{1}{x}}{-\frac{2}{x^3}} = \lim_{x \to 0^+} \frac{1}{x} \times -\frac{x^3}{2}$$

$$= 0$$

Ans: no: 03

3/(a).

Rate of change at $t = 2$ hours

$$C'(t) = 150(-0.12)e^{-0.12t} = -18e^{-0.12t}$$

$$C'(2) = -18e^{-0.12 \times 2} \approx -14.16 \text{ mg/L per hour}$$

$$C'(2) \approx -14.16 \text{ mg/L/hr}$$

(concentration is falling.)

A)

3/(b)

Time for concentration to drop to 30 mg/L

$$150e^{-0.12t} = 30 \Rightarrow e^{-0.12t} = 0.2$$

$$\Rightarrow -0.12t = \ln(0.2)$$

$$\Rightarrow t = \frac{\ln(0.2)}{-0.12} \text{ hours}$$

$$= 13.41 \text{ hours}$$ Ans

3/(c)

Average rate of change over $[0, 4]$

$$\left[C(4) - C(0)\right]/4 = \left[150e^{-0.48} - 150\right]/4$$

$$\approx -14.30 \text{ mg/L per hour}$$

3/(d)

$$\lim_{t\to\infty} \frac{C(t)+t}{3t+2} \qquad \left[\frac{\infty}{\infty}\right]$$

$$= \lim_{t\to\infty} \frac{C'(t)+1}{3}$$

$$= \lim_{t\to\infty} \frac{-18e^{-0.12t}+1}{3}$$

$$= \frac{0+1}{3} = \frac{1}{3}$$

Interpretation:

As $t\to\infty$ (the drug has essentially left the bloodstream ($C(t)\to 0$); so it contributes nothing to the ratio; the long run value is governed purely by the linear terms t and $3t$; giving $\frac{1}{3}$.

Ans: no: 04

$f(x) = x^4 - 4x^3 + 10$ on (∞, ∞).

4/(a)

Critical points:

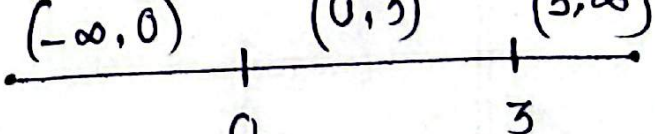
$$f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$$

$$f'(x) = 0 \Rightarrow 4x^2(x-3) = 0$$

$$\therefore x = 0, 3 \quad \text{(Ans)}$$

4/(b)

Since, $x^2 \geq 0$ always, the sign of $f'(x)$ is controlled entirely by the factor $(x-3)$.



- $(-\infty, 0)$: $f'(x)$ test: $f'(-1) < 0 \rightarrow$ decreasing.
- $(0, 3)$: $f'(x)$ test:

$f'(1) < 0 \longrightarrow$ decreasing

- $(3, \infty)$: $f'(x)$ test

$f'(4) > 0 \longrightarrow$ increasing

f is decreasing on $(-\infty, 3)$ and increasing on $(3, \infty)$

$$0 = 0$$

First Derivative Test:

- $x = 0$: f' is negative on both sides
- $x = 3$: f' changes from negative to positive

$\longrightarrow$ local minimum, $f(3) = 81 - 108 + 10 = -17$

$\therefore$ Local minimum at $(3, -17)$

$x = 0$ is neither a max nor a min

Ans:no:05

$$g(x) = 3x^5 - 5x^1 + 2$$

$$g'(x) = 15x^4 - 20x^3$$

$$g''(x) = 60x^3 - 60x^2 = 60x^2(x-1)$$

$x^2 \geq 0$ always; so the sign entirely depends on $(x-1)$

$(-\infty, 0)$ | $(0, 1)$ | $(1, \infty)$

0 1

- $(-\infty, 0)$: $g''(x) < 0$ ⟶ concave down
- $(0, 1)$: $g''(x) < 0$ ⟶ concave down
- $(1, \infty)$: $g''(x) > 0$ ⟶ concave up

∴ Concave down on $(-\infty, 1)$; concave up $(1, \infty)$

5/(b)

$g''(x) = 0$ at $x = 0$ & $x = 1$; but concavity only actually changes sign at $x = 1$; so there is a single inflection point:

$$g(1) = 3(1) - 5(1) + 2$$

$$= 0$$

$\therefore$ inflection point $(1, 0)$

5/(c)

Second Derivative Test:

$$g'(x) = 15x^4 - 20x^3$$
$$= 5x^3(3x - 4)$$

$\therefore g'(x) = 0$

$\Rightarrow 5x^3(3x-4) = 0$

$\therefore x = 0 ; \frac{4}{3}$ [critical points]

$g'(1) = 15 - 20 = -5 \neq 0$

$\therefore x = 1$ is not actually a critical point; it is the inflection point found in part (b).

Apply the second derivative test on the real critical points:

- $x = 4/3 : g''(4/3) = 60 \times \frac{16}{9} \times \left(\frac{4}{3} - 1\right)$
$= \frac{960}{27} > 0 \rightarrow$ local minimum

• $x = 0 : g''(0) = 0 \rightarrow$ test is inconclusive; the First Derivative Test shows $g' < 0$ on both sides of 0; so $x = 0$ is neither a max nor a min.